

Clustering Diffusion Model With Frequency-Signal Modulation for Variational Graph Autoencoders

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Abstract—Variational autoencoders (VAEs) have been widely used for node clustering, with existing methods mainly focusing on enhancing the expressiveness of their latent space. Recently, the integration of diffusion models with VAEs has provided new opportunities to achieve this objective. However, the mechanism by which the diffusion model improves performance remains unclear. To bridge this gap, we conduct an empirical analysis from the perspective of graph spectral theory, revealing that the signal modulation induced by diffusion models closely aligns with the low-frequency spectral characteristics of VAEs, which in turn explains their effectiveness. Nevertheless, further experiments highlight that diffusion models exhibit limitations in modulating high-frequency signals, which diverge from the spectral characteristics of VAEs. Moreover, existing diffusion methods fail to enable the latent space to adequately capture and reflect cluster-specific characteristics. To address these challenges, we propose a novel plug-and-play method, FVD, to improve the performance of VAE-based methods in node clustering tasks. Specifically, we incorporate the graph wavelet transform as a secondary signal modulator, enabling independent adjustments of specific frequency bands to better align with the spectral characteristics of VAEs. Additionally, we introduce the Student's t-distribution as a conditional constraint in the reverse process of FVD, deriving a more compact variational lower bound. This enhancement preserves fine-grained node information while focusing on clustering details, effectively mitigating the cluster collapse phenomenon. Comprehensive experimental results demonstrate that integrating FVD with existing methods achieves competitive performance improvements in most cases.

Index Terms—Variational graph autoencoder, node clustering, diffusion model, graph signal.

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The code of FVD is available at <https://github.com/Roiko97/FVD>.

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I. INTRODUCTION

NODE clustering (also known as community detection) is a fundamental task in the field of graph machine learning. It aims to partition nodes into distinct clusters while ensuring high intra-cluster compactness and clear inter-cluster separation [1]. Identifying cluster structures within networks facilitates a deeper understanding of node relationships, which support applications such as anomaly detection [2], [3] and drug discovery [4], [5]. High-performance node clustering methods largely relies on specially designed optimization functions. In this context, Variational Autoencoders (VAEs) [6], as generative methods, offer a natural optimization objective well suited for this task.

The probabilistic framework of traditional VAEs is illustrated in Fig. 1(a). Building upon this framework, existing VAE-based methods can be classified into two main groups. The first group extends the inference and/or generation processes by incorporating higher-order structural information, such as modularity [7], motifs [8] and k-core structures [9]. The second group introduces constraints to modify the optimization function [10], [11], [12]. In general, these methods primarily aim to improve the representational capacity of the latent space. Notably, recent studies have proposed combining Denoising Diffusion Probabilistic Model (DDPM) [13] (pipeline in Fig. 1(b)) with VAEs, which opens up new pathways to achieve this objective. For example, [14] propose to apply DDPM across multiple VAEs to support user recommendation. [15] design a sandwich-structured method that integrates variational autoencoder and DDPM for graph augmentation. Likewise, [16] incorporate DDPM into the latent space to guide the aggregation process of neighborhood information for anomaly detection.

Although these studies suggest that DDPM can effectively enhance the representational capacity of the latent space in variational graph autoencoder-based methods, the underlying mechanisms remain unclear. Against this backdrop, two questions arise: *What mechanism enables DDPM to improve the expressiveness of the latent space in VAEs? Can this mechanism help the latent space better reflect cluster characteristics?*

Essentially, noise can be regarded as a high-frequency signal within the graph spectrum. Consequently, the pipeline of DDPM can be understood as modifying specific frequency components in the graph, thereby altering the amplitude in the spectrum [17], [18]. This natural and intuitive connection motivates us to explore how DDPM affects latent representations from the spectral perspective. To this end, we begin with an empirical study (detailed in Section III-A) using a traditional VAE-based method, VGAE, to explore the spectral characteristics of VAEs.

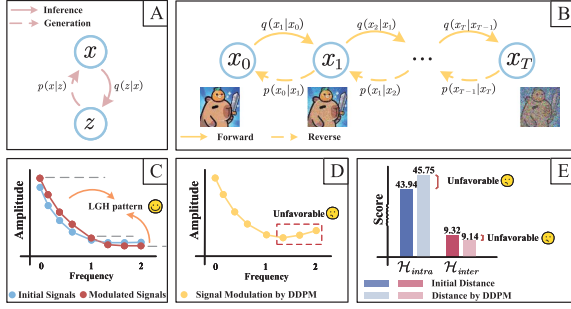


Fig. 1. (a) is the probabilistic framework of VAEs. (b) presents the pipelines of DDPM. (c) highlights the spectral characteristics of VAEs for low-frequency signals, while intentionally suppressing high-frequency components during signal modulation is needed. (d) depicts the signal modulation results of DDPM, indicating a deviation from the LGH pattern in the high-frequency band. (e) outlines the detrimental effects of DDPM on intra-cluster compactness and inter-cluster separation.

Our observations reveal that VGAE prioritizes low-frequency signals, indicating a clear inclination towards preserving their features, while high-frequency signals receive relatively less emphasis. Based on this, we summarize a general mathematical pattern, called LGH, for VAEs from the spectral perspective, as shown in Fig. 1(c): **after modulation, the integral of amplitude difference function in the Low-frequency band should be Greater than zero, while the integral in High-frequency band should be less than zero.**

Within the constraints of the LGH pattern, we adopt two more comprehensive experimental designs to discuss the two questions raised earlier. Specifically, we design the second empirical experiment to evaluate how well the signal modulation results produced by DDPM align with the LGH pattern (detailed in Section III-B). Additionally, we conduct a third experiment to assess the impact of DDPM on clustering characteristics (detailed in Section III-C).

- *Finding (1):* We observe that the signal modulation results of DDPM are generally consistent with the signal characteristics required by VAEs, which explains why DDPM enhances the expressiveness of the latent space. However, VAEs exhibit a progressively diminishing response to frequency signals as the frequency increases from low to high. This behavior indicates that amplifying low-frequency signals or suppressing high-frequency signals can lead to performance gains, or vice versa. However, as visualized in Fig. 1(d), DDPM amplifies signals across the entire frequency band, which differs from the requirement of VAEs for high-frequency signals.
- *Finding (2):* A well-structured cluster should exhibit high intra-cluster compactness and inter-cluster separation. However, DDPM amplifies signals across all frequency bands, which may lead to the loss of key structural information related to clustering. This may lead to node escape or the formation of ambiguous clusters, a phenomenon we term as cluster collapse. Specifically, as outlined in Fig. 1(e), after signal modulation, we observe that intra-cluster compactness and inter-cluster separation may adjust in the opposite direction of the expected results.

According to the aforementioned findings and analysis, we propose a cluster-driven Diffusion model that integrates Frequency signal modulation (FVD). As a plug-and-play method, it improves the performance of existing VAE-based methods without modifying their core structures. Specifically, we introduce graph wavelet transform as an adapter for the DDPM to reduce its coupling with the latent space. Meanwhile, secondary signal modulation through graph wavelet transform ensures that the modulated signals conform to the LGH pattern. To address cluster collapse, inspired by [12], we incorporate the Student's t-distribution [19] as a conditional constraint into reverse process of DDPM. At each reverse diffusion step, noise removal occurs under the guidance of cluster information, thus preventing the misinterpretation of boundary and intra-cluster differences as noise. Our contributions are summarized as follows:

- *Analysis:* We investigate the spectral behavior of VAEs and propose the LGH pattern to characterize it. Building on this foundation, we explain how DDPM enhances the representational capacity of the latent space in VAEs, while also elucidating its limitations.
- *Method:* (1) We propose a plug-and play method FVD, which integrates graph wavelet transform module as an adapter to ensure modulated signals conform to the LGH pattern, while independent secondary signal modulation addresses the bottleneck caused by the strong coupling between DDPM and the latent space. (2) We develop a conditional diffusion module to mitigate cluster collapse by designing a novel reverse process and deriving a variational lower bound, effectively preserving cluster boundaries and intra-cluster differences. To the best of our knowledge, this is the first application of a conditional diffusion method to node clustering with VAEs.
- *Experiments:* We apply FVD to eight representative VAE-based methods for node clustering. Experimental results on five benchmark datasets demonstrate that integrating FVD enables these methods to achieve competitive performance improvements while maintaining their original module structures unchanged.

II. PRELIMINARIES

A. Node Clustering

An undirected graph $G = (\mathbf{A}, \mathbf{X})$ comprises interactions among N nodes, where $\mathbf{A} \in \mathbb{R}^{N \times N}$ represents the adjacency matrix and $\mathbf{X} \in \mathbb{R}^{N \times M}$ denotes the attribute matrix. For any pair of nodes u and v , if there is a interaction between them, $\mathbf{A}_{u,v} = 1$, otherwise, $\mathbf{A}_{u,v} = 0$. Each node is associated with an M -dimensional attribute vector that represents its inherent features.

The graph clustering task utilizes node information to partition them into k disjoint clusters $C = \{C_1, C_2, \dots, C_k\}$ in an unsupervised manner, where C_k is the k -th cluster.

B. Variational Autoencoder

VAE-based methods primarily consist of inference, generation and the Evidence Lower Bound (ELBO), as illustrated in the Fig. 1(a). Specifically, for an input sample x , the true

posterior distribution $p(z)$ is unknown. To approximate $p(z)$, a common approach is to introduce the variational distribution $q(z|x) \sim \mathcal{N}(\mu, \sigma^2)$ to infer z , and the observation distribution $p(x|z) \sim \text{Bernoulli}(x)$ to generate x . By maximizing ELBO, the inference and generation processes progressively capture the structural and distributional characteristics of the data, thereby approximating the true representation distribution. Without loss of generality, the ELBO is derived using Bayes' theorem as follows [6]:

$$\begin{aligned} \log p(x) &\geq \int_z q(z|x) \log p(x|z) dz - KL(q(z|x)||p(z)) \\ &= \int_z q(z|x) \log p(x|z) dz - \frac{1}{2}(\mu^2 + \sigma^2 - \log \sigma^2 - 1) \end{aligned} \quad (1)$$

where the first term can be directly computed via binary cross-entropy, KL represents the Kullback-Leibler divergence, and $p(z) \sim \mathcal{N}(0, 1)$. Finally, the learned node representations z are clustered using either K-means or an *argmax*-based strategy. For more details, please refer to Section VI-A.

C. Denoising Diffusion Probabilistic Model

The DDPM pipeline, as shown in Fig. 1(b), which consists of two main processes. Given an input sample $x_0 \sim p(x)$, the forward process involves gradually injecting Gaussian noise into the sample via a Markov chain until its structure is completely lost. Formally, let T denote the total number of steps. For any noisy sample generated at step t , its distribution is $p(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t\mathbf{I})$, where $t \in \{1, 2, \dots, T\}$ and β_t is a hyperparameter that controls the noise scale at step t . As $T \rightarrow +\infty$, x_T approaches $\mathcal{N}(0, 1)$ [13]. In the reverse process, noise is progressively removed, ultimately recovering the original sample x_0 . Mathematically, given a noisy sample x_t , the denoising process is defined iteratively by $q_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_\theta(x_t, t))$, where θ is learned via a neural network. With the maximizing of variational lower bound (VLB) on log-likelihood, DDPM identifies and removes noise to generate a new sample. Without loss of generality, the VLB is derived using Bayes' theorem and the reparameterization trick as follows:

$$\log p(x) \geq \mathbb{E}_{q(x_1, \dots, x_T|x_0)} \log \frac{\prod_{t=1}^T p(x_t|x_{t-1})}{q_\theta(x_T) \prod_{t=1}^T q_\theta(x_{t-1}|x_t)} \quad (2)$$

D. Graph Spectrum Theory

Graph spectrum theory is a fundamental concept in graph signal processing. It relies on the eigenvalues and eigenvectors of the graph Laplacian matrix $\mathbf{L} = \mathbf{D} - \mathbf{A}$ to modulate node signals, where $\mathbf{D}$ is node degree matrix. $\hat{\mathbf{L}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$ is the symmetric normalized graph Laplacian matrix. Through eigendecomposition, it can be decomposed into the product of three matrices, i.e., $\hat{\mathbf{L}} = \mathbf{U}^T \Lambda \mathbf{U}$, where $\mathbf{U} = [u_1, u_2, \dots, u_N]$ is an orthonormal eigenmatrix, and $\Lambda = \text{diag}[\lambda_1, \lambda_2, \dots, \lambda_N]$ is a diagonal matrix. For each $\lambda_i \in [0, 2]$ arranged in ascending order, there is a corresponding eigenvector u_i .

In graph signal processing, low-frequency signals $\mathcal{F}_-$ represent smooth information among nodes and correspond to

eigenvectors associated with eigenvalues ranging from λ_1 to $\lambda_{N/2}$ ($\lambda_{N/2} \approx 1$). In contrast, high-frequency signals $\mathcal{F}_+$ represent non-smooth information among nodes and are generally considered noise. They correspond to eigenvectors associated with eigenvalues ranging from $\lambda_{(N/2)+1}$ to λ_N ($\lambda_N \approx 2$) [20], [21]. Amplitude $\mathcal{A}$ quantifies the strength of each frequency signal, and is derived by applying the Fourier transform [22], i.e., $\mathcal{A} = \mathbf{U}^T z$.

III. EMPIRICAL STUDY

In our empirical study, we mainly explore three questions: (Q1) What characteristics do the VAEs exhibit for frequency signals? (Q2) What mechanism enables DDPM to enhance the expressiveness of the latent space in VAEs? Furthermore, (Q3) Can this mechanism help the latent space better reflect cluster characteristics?

A. (Q1) What Characteristics Do the VAEs Exhibit for Frequency Signals?

Experimental design: Given a symmetric normalized graph Laplacian matrix $\hat{\mathbf{L}}$, it can be rewritten via eigenvalue decomposition as follows:

$$\hat{\mathbf{L}} = \underbrace{(\lambda u u^T)_1 + \dots + (\lambda u u^T)_{N/2}}_{\mathcal{F}_-} + \underbrace{(\lambda u u^T)_{N/2+1} + \dots + (\lambda u u^T)_N}_{\mathcal{F}_+} \quad (3)$$

To verify the characteristics of VAEs for different frequency signals, we discard signals in two directions at ratios of $\{0\%, 5\%, 10\%, 15\%, 20\%, 25\%\}$. Fig. 2(a) illustrates this process. Specifically, we progressively discard signals from $\lambda = 0$ toward $\lambda = 1$ according to the specified ratios, while retaining the remaining components unchanged. For example, discarding the corresponding frequency signal at a rate of 5%, the reorganized $\hat{\mathbf{L}}$ is as follows:

$$\hat{\mathbf{L}} := (\lambda u u^T)_{N \times 5\%+1} + \dots + (\lambda u u^T)_{N/2} + \mathcal{F}_+ \quad (4)$$

Similarly, we perform the same procedure at $\lambda = 2 \rightarrow \lambda = 1$. Given the foundational role of VGAE in VAEs, we select it as a representative method and conduct experiments on the Cora dataset. Specifically, we replace the symmetric normalized adjacency matrix in the inference phase with the reorganized $\hat{\mathbf{L}}$ as the input for graph convolution. This design is motivated by the fact that signal propagation in graph convolution essentially corresponds to spectral filtering via the graph Laplacian. Therefore, using reorganized $\hat{\mathbf{L}}$ can be regarded as applying a spectral modulation kernel, enabling us to explore how the model responds to different spectral configurations and to analyze its frequency characteristics. We evaluate its performance on node clustering tasks using Accuracy (ACC), Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), and F1-macro (F1).

Results and analysis: Fig. 3 presents the performance results of VGAE when signals are randomly removed at various ratios in two different directions. We find that even removing a small portion of the lowest-frequency signals significantly degrades

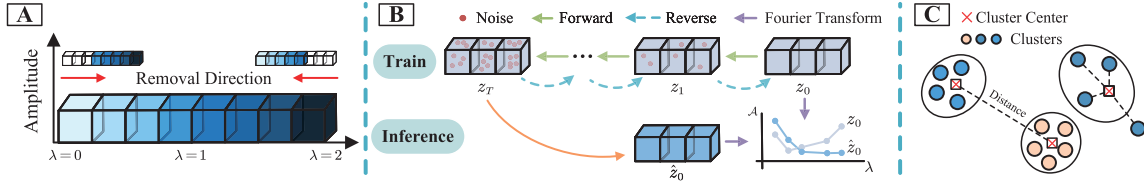


Fig. 2. (a) is the first empirical study, designed to illustrate the frequency characteristics of VAEs. (b) corresponds to the second empirical study, focusing on the modulation effects of DDPM on frequency signals. (c) explore the third empirical study, which aims to evaluate the impact of DDPM on intra-cluster compactness and inter-cluster separation.

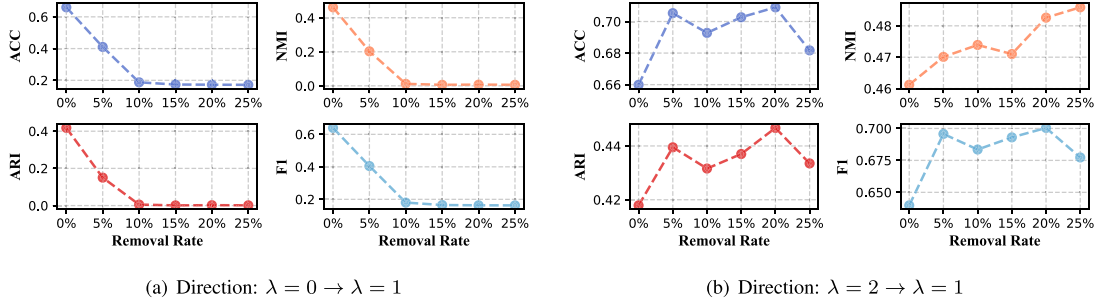


Fig. 3. VGAE performance changes under different signal removal rates.

VGAE performance. As shown in Fig. 3(a), all evaluated metrics decline noticeably, among which key clustering metrics NMI and ARI approaching zero. In contrast, discarding a small portion of the highest-frequency signals can yield substantial performance gains. As depicted in Fig. 3(b), applying any removal rate above 0% leads to measurable improvements compared to retaining all signals.

From the spectral perspective, low-frequency components mainly capture node similarity, while high-frequency components emphasize differences. Since clustering naturally relies on aggregating similar nodes, suppressing low-frequency signals undermines the similarity characteristics essential for effective clustering. Meanwhile, removing the high-frequency signals reduces their distinguishing differences. Both adjustments can lead to pronounced changes in the observed outcomes. Moreover, frequency signals near $\lambda = 1$ often carry information about cluster boundaries [23], with the left side generally exhibiting smoother boundary details compared to the right. This is because low-frequency signals tend to capture globally smooth structures, while high-frequency components reflect sharp local variations. In contrast, transitions near cluster boundaries typically exhibit intermediate variations. Therefore, removing these smoother details may disrupt the structural integrity of the clusters, causing their boundaries to blur and stack. Conversely, filtering out less smooth boundary information can help reveal the boundaries more clearly.

Based on the above results and findings, we describe the spectral characteristics of VAEs as follows: **(A1) The latent representation of VAEs tends to capture low-frequency information while attenuating high-frequency components:** Based on this characteristic, we summarize a general pattern LGH, for VAEs from the spectral perspective: after modulation, the integral of amplitude difference function in the Low-frequency band should be Greater than zero, while the integral in High-frequency band should be less than zero.

Mathematically, the LGH pattern is defined as:

$$\int_0^1 \Phi(\mathcal{A}) d\mathcal{A} > 0 > \int_1^2 \Phi(\mathcal{A}) d\mathcal{A} \quad (5)$$

where $\Phi(\mathcal{A}) = \mathcal{A} - \mathcal{A}_0$ is amplitude difference function and $\mathcal{A}_0$ is the initial amplitude. Note that this pattern only constrains the amplification and attenuation of the overall signal in the low and high frequency bands, without imposing further constraints on the internal details of these bands.

B. (Q2) What Mechanism Enables DDPM to Enhance the Expressiveness of the Latent Space in VAEs?

Experimental design: Given that the processing of noise by DDPM can be viewed as the modification of specific frequency components, we explore its underlying mechanism in the latent space from the spectral perspective. We choose node2vec [24] to generate the input representation z_0 for DDPM instead of adopting the output from VAEs. The main reason is that prior work [25] often employs node2vec to generate missing node attributes. Following this strategy helps reduce input-specific dependencies (e.g., VAE outputs) and thereby minimizes potential bias when studying the spectral characteristics of DDPM. Fig. 2(b) shows this process. Specifically, we train DDPM for 200 epochs on the Cora, Citeseer and Pubmed datasets, respectively. Once trained, new samples $\hat{z}_0$ are inferred by sampling noisy data from z_T and iteratively denoising. Finally, we map the representations z_0 and $\hat{z}_0$ to the spectral domain using Fourier transform to observe the amplification and attenuation of amplitudes across different frequency bands by DDPM.

Results and analysis: Fig. 4 shows the amplitude magnification ratio across different frequency bands before and after applying DDPM, computed as $(\mathcal{A}_{\hat{z}_0} - \mathcal{A}_{z_0}) / \mathcal{A}_{z_0}$. We observe that DDPM can be viewed as a signal modulator that amplifies the most signal across the entire frequency band. Specifically, we

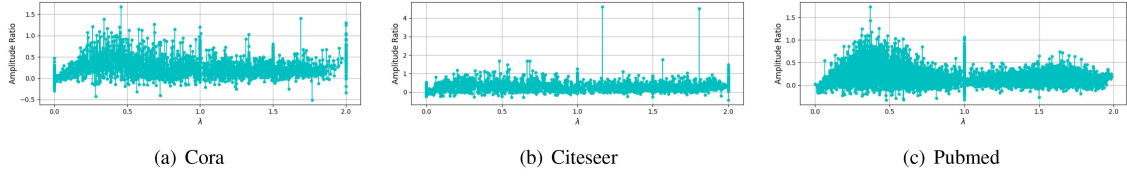


Fig. 4. Amplification ratio of signal amplitude before and after DDPM modulation. After applying DDPM, 92.95%, 92.31% and 92.39% of the nodes in Cora, Citeseer and Pubmed, respectively, exhibit increased signal amplitudes.

TABLE I
EVALUATING THE INFLUENCE OF DDPM ON INTRA-CLUSTER COMPACTNESS AND INTER-CLUSTER SEPARATION. BOTH ▼ AND ▲ REPRESENT POSITIVE RESULTS.

Methods	node2vec	DDPM	node2vec	DDPM
Metrics	$\mathcal{H}_{intra}$		$\mathcal{H}_{inter}$	
Cora	41.55	34.50▼	8.52	9.98▲
Citeseer	43.94	45.75	9.31	9.14
Pubmed	63.07	63.67	28.17	26.60

observe that 92.95%, 92.31% and 92.39% of the nodes in Cora, Citeseer and Pubmed, respectively, experience an increase in their signal amplitude after DDPM modulation. Furthermore, the amplification effect in the low-frequency range is more pronounced compared to the high-frequency range, which is consistent with the spectral characteristics of VAEs in the low-frequency band described in (A1). Notably, a representative study [26] in graph signal processing demonstrates that fully leveraging the implicit structural information in signals can significantly enhance the expressiveness of latent representations. Inspired by this finding, and in conjunction with the spectral characteristics exhibited by VAEs, we propose that (A2) **DDPM enhances the latent space expressiveness of VAEs mainly by amplifying low-frequency signals**.

However, DDPM does not fully follow the LGH pattern for high-frequency signals. Specifically, the modulation of frequency signals by DDPM not only amplifies low-frequency signals but also enhances high-frequency signals, sometimes even emphasizing the highest frequency components (as shown in Fig. 4(c)), which contradicts spectral characteristics of VAEs for high-frequency signals. Misalignment in signals can impair the reconstruction capability of VAEs during decoding, a phenomenon also observed in spectral analysis of DDPM within the vision domain [27]. Therefore, we need to consider how to adjust the frequency modulation results of DDPM to ensure alignment with the LGH pattern. This will be discussed in Section IV-D.

C. (Q3) Can This Mechanism Help the Latent Space Better Reflect Cluster Characteristics?

Experimental design: Based on the node representations z_0 and $\hat{z}_0$ obtained in (Q2), we compute the intra-cluster compactness $\mathcal{H}_{intra}$ and inter-cluster separation $\mathcal{H}_{inter}$ as follows:

$$\mathcal{H}_{intra} = \max_{c_1, c_2, \dots, c_k} \frac{1}{N_k} \sum_{i \in c_k} \|z_i - \mu_k\|^2 \quad (6)$$

$$\mathcal{H}_{inter} = \min_{i, j \in C, i \neq j} \|\mu_i - \mu_j\| \quad (7)$$

where N_k is the number of nodes in the cluster k , and μ is the cluster center. $\mathcal{H}_{intra}$ quantifies the interaction density within a cluster, where a smaller value indicates greater internal cohesion. $\mathcal{H}_{inter}$ measures the separation between communities, where a larger value indicates clearer segregation between communities. Fig. 2(c) visualizes the above calculation.

Results and analysis: Table I presents the results of the calculations for $\mathcal{H}_{intra}$ and $\mathcal{H}_{inter}$. We observe that, in most cases, the representation after frequency modulation using DDPM leads to slight dispersion of nodes within clusters and a reduction in inter-cluster distance. We propose two possible explanations for this. First, the significant amplification of low-frequency signals by DDPM indicates its focus on capturing individual details during the reverse process. While this may enhance fine-grained details, it can also lead to the loss of key structural information related to clusters. Building on this, DDPM amplifies high-frequency components that reflect non-smooth information. This amplification may cause nodes to cross cluster boundaries (as shown in Fig. 2(c)), resulting in node escape or even cluster stacking. We refer to this phenomenon as cluster collapse. To this end, we argue that (A3) **the modulation of signals by DDPM does not enable the latent space to reflect clustering characteristics well**. This motivates us to explore an approach for improving DDPM that prevents the loss of cluster structural features during the reverse diffusion process, thereby alleviating cluster collapse. This will be discussed in Section IV-C.

IV. METHODOLOGY

A. Overview

Two key findings from empirical experiments deserve attention and are summarized as follows:

- DDPM functions as a signal modulator that is capable of modulating the frequency signals within the latent space of VAEs. However, its modulation of high-frequency signals contravenes the LGH pattern, thereby limiting the expressiveness of the latent space in VAEs.
- DDPM exhibits a tendency to focus on fine-grained details, yet this often neglects cluster information with localized characteristics. With the amplification of high-frequency signals, the differences among nodes are further emphasized, making clusters more likely to collapse.

For the first problem, a straightforward approach is to repeatedly adjust the parameters in DDPM (e.g., diffusion step, noise scale, etc. [13]) to minimize noise during the reverse diffusion process, thus attenuating high-frequency signals. However, this black-box approach leads to modulated results that lack controllability. Even when some degree of control is achieved, the input

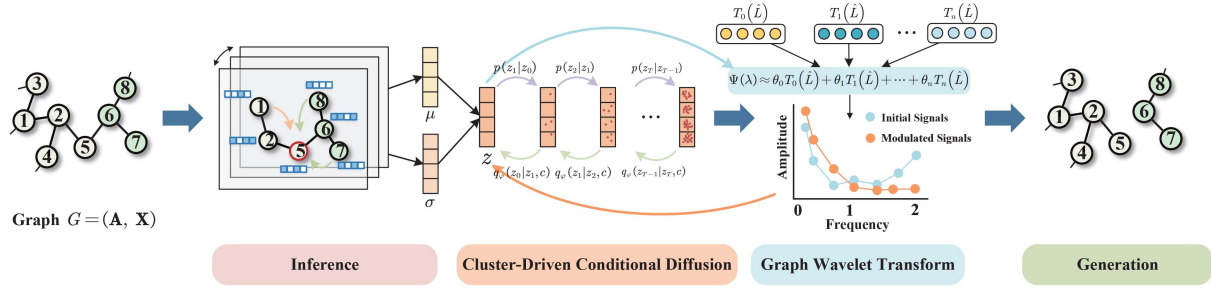


Fig. 5. The framework of FVD. Starting with graph information G , the inference component of VAE projects this data into a latent space to learn the node embedding z . The conditional diffusion module then processes z by iteratively adding and removing noise, modulating its frequency signals to better capture clustering structures. Following this, the graph wavelet transform applies independent secondary signal modulation, aligning the final frequency signals with the LGH pattern. Finally, the generation component of the VAE reconstructs the original graph structure.

to DDPM is dependent on the output of encoder, which in turn is determined by the design of model. This strong coupling results in the inevitability of extensive manual tuning, which limits the generalizability of the model. As for the second issue, existing methods [11], [28] attempt to address the inherent limitations of VAEs to mitigate cluster collapse. However, these methods cannot be directly extended to DDPM. Moreover, for DDPM variants such as conditional diffusion models, the focus has predominantly been on the visual domain, with limited exploration in graph-related tasks. To the best of our knowledge, no work has yet applied conditional diffusion methods to graph clustering tasks.

To address the aforementioned issues, and based on the conclusions drawn from our empirical study, we propose a frequency-signal modulated cluster-driven diffusion model (FVD). As a plug-and-play method, FVD enhances the performance of existing VAE frameworks without modifying their core structures. Specifically,

- To ensure that signal modulation by DDPM aligns with the LGH pattern while minimizing strong coupling with the integrated method, we introduce the graph wavelet transform for independent secondary modulation. Building on the amplification of low-frequency signals by DDPM, the graph wavelet transform enable fine-tuning within this frequency band while effectively suppressing high-frequency components, thereby ensuring alignment with the spectral characteristics of VAEs.
- To address the issue of cluster collapse, we introduce the Student's t-distribution as a constraint within the reverse diffusion process. At each denoising step, DDPM not only captures fine-grained details but also emphasizes the preservation of local cluster information. Additionally, with the attenuation of high-frequency signals, inherent differences at cluster boundaries and within clusters are preserved, rather than being mistakenly removed as noise.

Fig. 5 presents the framework of FVD. Given graph information G , the inference component of VAE maps this data into the latent space to learn the node embedding z . Within the conditional diffusion module, the latent representation z undergoes iterative noise addition and denoising, modulating its frequency signals to capture clustering structure information. Subsequently, graph wavelet transform performs independent

secondary signal modulation, ensuring that the final frequency signals follow the LGH pattern. Finally, the generation component of VAE reconstructs the original graph. The model is trained by iteratively optimizing the ELBO and VLB to complete the graph clustering task. In the next section, we will provide a step-by-step introduction to the FVD pipeline.

B. Inference Component of VAEs

Consider a node x randomly drawn from the sample space. In general, x is usually considered to follow a Gaussian distribution to infer the latent representation z [29]. Formally, the representation z obtained by the reparameterization trick can be written as:

$$z = \mu + \epsilon\sigma \quad (8)$$

where ϵ is sampled from $\mathcal{N}(0, 1)$. Without loss of generality, let $f(x)$ denote the node-related information associated with x . The parameters μ and σ are then computed by two-layer Graph Convolutional Network (GCN):

$$\mu = \text{COV}_{\phi_\mu}(\text{COV}_\phi(f(x)), f(x)) \quad (9)$$

$$\sigma = \text{COV}_{\phi_\sigma}(\text{COV}_\phi(f(x)), f(x)) \quad (10)$$

where ϕ_μ , ϕ_σ and ϕ are learnable parameters of GCN. COV denotes convolution operation.

C. Cluster-Driven Conditional Diffusion Modules

Forward process: Vanilla DDPM progressively adds Gaussian noise to the node x through a Markov chain during forward process, gradually destroying its structure. At each step, DDPM generates a sequence of perturbed sample $\{x_1, x_2, \dots, x_T\}$, with the perturbations becoming more obvious. The toy image in the Fig. 1(b) shows illustrates this process. The forward process of our conditional diffusion module adopts the same strategy. Specifically, given a node representation $z_0 = z$ obtained from the output of VAEs, the distribution of a sequence of perturbed samples $\{z_1, z_2, \dots, z_T\}$ is expressed as $p(z_1, z_2, \dots, z_T|z_0)$. According to the Markov chain, this distribution can be rewritten as:

$$p(z_{1:T}|z_0) = \prod_{t=1}^T p(z_t|z_{t-1}) \sim \mathcal{N}(z_t|\sqrt{1-\beta_t}z_{t-1}, \beta_t\mathbf{I}) \quad (11)$$

where $\beta_t \in (0, 1)$ controls the scale of the Gaussian noise added at each step t .

Based on (11), the properties of Gaussian distribution and the reparameterization trick, we derive the conclusion that the perturbation distribution at any given step can be fully determined by z_0 and β_t . Specifically, we can obtain the distribution of z_t and z_{t-1} via (11) and further rewrite them via the reparameterization trick as:

$$z_t = \sqrt{\alpha_t} z_{t-1} + \sqrt{1 - \alpha_t} \epsilon_{t-1} \quad (12)$$

$$z_{t-1} = \sqrt{\alpha_{t-1}} z_{t-2} + \sqrt{1 - \alpha_{t-1}} \epsilon_{t-2} \quad (13)$$

where $\alpha_t = 1 - \beta_t$.

By substituting (13) into (12), we can rewrite (12) as follows:

$$\begin{aligned} z_t &= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} z_{t-2} + \sqrt{1 - \alpha_{t-1}} \epsilon_{t-2}) + \sqrt{1 - \alpha_t} \epsilon_{t-1} \\ &= \sqrt{\alpha_t \alpha_{t-1}} z_{t-2} + \sqrt{\alpha_t - \alpha_t \alpha_{t-1}} \epsilon_{t-2} + \sqrt{1 - \alpha_t} \epsilon_{t-1} \\ &\because \sqrt{\alpha_t - \alpha_t \alpha_{t-1}} \epsilon_{t-2} + \sqrt{1 - \alpha_t} \epsilon_{t-1} \sim \mathcal{N}(0, 1 - \alpha_t \alpha_{t-1}) \\ &\therefore z_t = \sqrt{\alpha_t \alpha_{t-1}} z_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \epsilon \end{aligned} \quad (14)$$

By iteratively applying the above equation until $t = 0$, we obtain:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad (15)$$

where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.

Overall, during the forward process, given the initial representation input z_0 and noise schedule β_t , the distribution of the any perturbation representation z_t can be expressed as:

$$q(z_t|z_0) \sim \mathcal{N}(z_t | \sqrt{\bar{\alpha}_t} z_0, 1 - \bar{\alpha}_t) \quad (16)$$

To regulate the noise added in $z_{1:T}$, we follow [14] to use a linear noise schedule for $1 - \bar{\alpha}_t$, i.e.,

$$1 - \bar{\alpha}_t = \zeta \cdot \left[\alpha_{\min} + \frac{t-1}{T-1} (\alpha_{\max} - \alpha_{\min}) \right] \quad (17)$$

where ζ is a parameter that controls the noise scale, and $\alpha_{\min} < \alpha_{\max} \in (0, 1)$ are the lower and upper bounds of the added noise.

Reverse process: Vanilla DDPM gradually removes noise from the sample x_T via a Markov chain during the reverse process, ultimately reconstructing the original data. However, as discussed in Section III-C, DDPM tends to overlook cluster information with localized characteristics, while inadvertently emphasizing node differences. In this context, the vanilla reverse diffusion approach tends to lead to cluster collapse. To mitigate this issue, we introduce the Student's t-distribution as a constraint in the reverse diffusion process to guide data recovery. While existing VAE-based clustering methods [11], [12], [29] employ Gaussian mixture models (GMM) to enable end-to-end clustering, each component in a GMM requires learning multiple parameters and is also prone to cluster collapse during pretraining. In contrast, the Student's t-distribution, with its heavy-tailed nature, is more robust to local structural outliers commonly observed in graph. This property allows it to better capture local and boundary cluster structures, thereby facilitating the preservation of cluster information during the reverse process. Formally, according to a Markov chain, the sample distribution

in the reverse process is:

$$q_\varphi(z_{0:T}) = q(z_T) \prod_{t=1}^T q_\varphi(z_{t-1}|z_t, c) \quad (18)$$

where φ is learnable parameter in any neural networks, and clustering constraint c is a learnable parameter, initialized by the cluster centers obtained from applying K-means to the node2vec-generated representations.

Assuming that z_t and c are independent, the conditional distribution in the above equation can be further approximated as:

$$q_\varphi(z_{t-1}|z_t) \sim \mathcal{N}(z_{t-1} | \mu_\varphi(z_t, t), \sigma_\varphi(z_t, t)) \quad (19)$$

$$q_\varphi(z_{t-1}|c) \sim \mathcal{N}(z_{t-1} | 0, \tilde{\sigma}(c, z_t)) \quad (20)$$

It is worth noting that as $T \rightarrow +\infty$, the distribution z_T will approximate the standard Gaussian distribution. To this end, the reverse diffusion process can be interpreted as a gradual adjustment of the mean and variance of this standard Gaussian. In this context, in (20), we assume that, under the cluster constraint c , the distribution of z_t remains Gaussian, with its variance determined by the Student's t-distribution:

$$\tilde{\sigma}(c, z_t) = \frac{(1 + \|z_t - \Theta_i\|^2)^{-1}}{\sum_{i'} (1 + \|z_t - \Theta_{i'}\|^2)^{-1}} \quad (21)$$

where Θ_i is the cluster center of the i -th cluster and is initialized by K-means.

D. Graph Wavelet Transform

Graph wavelet transform allows independent manipulation of different frequency components, enabling effective amplification or attenuation of specific frequency components [30], [31]. Building on the advantages of graph wavelet transform observed in related studies [23], [32], we adopt it as a secondary frequency modulation method to ensure that the final frequency-modulated signal adheres to the LGH pattern. This strategy ensures that the frequency signals modulated by DDPM remain stable in the low-frequency range while further attenuating high-frequency components. Similar to graph Fourier transform [33], components with smaller eigenvalues in graph wavelet transform correspond to low-frequency signals, while those with larger eigenvalues correspond to high-frequency signals. Mathematically, the normalized Laplacian matrix is symmetric and semi-positive definite. According to the theory of graph signal processing [34], its eigenvectors can be used as the basis for the graph wavelet transform. Specifically, given a signal-modulated representation z , the graph wavelet transform is defined as [35]:

$$\hat{z} = \Psi(\lambda) z = \mathbf{U} g_s(\lambda) \mathbf{U}^T z \quad (22)$$

where $g_s(\lambda) = \text{diag}(\exp(s\lambda_1), \exp(s\lambda_2), \dots, \exp(s\lambda_N))$ is a scaling matrix with heat kernel scaled by parameter s .

Notably, computing (22) results in the same computational burden as calculating the graph Fourier transform. Specifically, as discussed in Section II-D, computing the eigenvector $\mathbf{U}$ typically requires $O(N^3)$ time complexity, which becomes prohibitive for large graphs. Fortunately, existing research [35] has

demonstrated that the graph wavelet transform can be efficiently approximated using Chebyshev polynomials, reducing the time complexity to a linear relationship with the number of edges in the graph. Consequently, $\Psi(\lambda)$ can be rewritten as:

$$\Psi(\lambda) \approx \sum_{i=0}^K \theta_i \text{Poly}_i(\hat{\mathbf{L}}) \quad (23)$$

where $\text{Poly}_0(\hat{\mathbf{L}}) = \mathbf{I}$, $\text{Poly}_1(\hat{\mathbf{L}}) = \hat{\mathbf{L}}$, and $\text{Poly}_i(\hat{\mathbf{L}}) = 2\hat{\mathbf{L}}\text{Poly}_{i-1}(\hat{\mathbf{L}}) - \text{Poly}_{i-2}(\hat{\mathbf{L}})$. θ is the filter coefficient and K is the order of the polynomial.

E. Generation Component of VAEs

Given the representation z after graph wavelet transform, the generation component of the VAEs reconstruct the original graph as follows:

$$p(\hat{\mathbf{A}}_x = 1 | \hat{z}) = \varrho(\hat{z}\hat{z}^T) \quad (24)$$

where ϱ is a non-linear function, such as sigmoid function.

F. Optimization

VAEs optimization: To ensure that the prior distribution assumed by the VAEs gradually approximates the true posterior distribution, the training objective is to maximize the ELBO. For unconditional VAEs, the ELBO can be further rewritten via (1) as:

$$\begin{aligned} \mathcal{L}_{ELBO} = & \frac{1}{L} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N \mathbf{A}_{ij} \log \hat{\mathbf{A}}_{ij}^{(l)} \\ & + (1 - \mathbf{A}_{ij}) \log(1 - \hat{\mathbf{A}}_{ij}^{(l)}) \\ & - KL(q(\hat{\mathbf{Z}} | f(G)) || p(\hat{\mathbf{Z}})) \end{aligned} \quad (25)$$

where L is the number of samples from Monte Carlo Stochastic Gradient Variational Bayes estimator [6], and $\hat{\mathbf{Z}} = [\hat{z}_1, \hat{z}_2, \dots, \hat{z}_N]$. For details on the ELBO of conditional VAEs, refer to [12].

FVD optimization: To optimize FVD, the training objective is to maximize the VLB on the log-likelihood. However, due to the introduction of a cluster-driven constraint in the reverse process, the standard VLB (i.e., (2)) is insufficient for training FVD. Below, we will derive the new VLB for FVD.

Given the input representation z_0 , the t -th step representation z_t and the constraint c , the $(t-1)$ -th step representation z_{t-1} can be derived via Bayes's theorem:

$$p(z_{t-1} | z_t, z_0, c) = \frac{p(z_{t-1}, z_t, z_0, c)}{p(z_t, z_0, c)} \quad (26)$$

where joint probability distribution can be rewritten as:

$$p(z_{t-1}, z_t, z_0, c) = p(z_t | z_{t-1}, z_0, c) p(z_{t-1} | z_0, c) p(z_0 | c) p(c) \quad (27)$$

It is worth noting that the constraint c is implicitly introduced in the forward distribution to simplify the computation of the final VLB. In practice, whether the constraint c is explicitly

incorporated into the forward process does not fundamentally affect the distribution of the final representation z_T .

Since FVD follows Markov chains in both the forward and reverse processes, the above equation can be simplified to:

$$p(z_t | z_{t-1}, z_0, c) = p(z_t | z_{t-1}, c) \quad (28)$$

According to the independence assumption, (26) can be further approximated as:

$$p(z_{t-1}, z_t, z_0, c) = \frac{p(z_t | z_{t-1}) p(z_{t-1} | z_0) q(z_{t-1} | c)}{p(z_t | z_0)} \quad (29)$$

where

$$\begin{cases} p(z_t | z_{t-1}) \sim \mathcal{N}(z_t | \sqrt{\alpha_t} z_{t-1}, 1 - \alpha_t) \\ p(z_t | z_0) \sim \mathcal{N}(z_t | \sqrt{\alpha_t} z_0, 1 - \alpha_t) \\ p(z_{t-1} | z_0) \sim \mathcal{N}(z_{t-1} | \sqrt{\alpha_{t-1}} z_0, 1 - \alpha_{t-1}) \\ q(z_{t-1} | c) \sim \mathcal{N}(z_{t-1} | 0, \tilde{\sigma}) \end{cases} \quad (30)$$

Based on the probability density function of the Gaussian distribution, (29) can be expanded as:

$$\begin{aligned} p(z_{t-1}, z_t, z_0, c) & \propto \exp \left\{ -\frac{1}{2} \left[\frac{(z_t - \sqrt{\alpha_t} z_{t-1})^2}{1 - \alpha_t} \right. \right. \\ & \quad \left. \left. + \frac{(z_{t-1} - \sqrt{\alpha_{t-1}} z_0)^2}{1 - \alpha_{t-1}} + \frac{(z_{t-1})^2}{\tilde{\sigma}} - \frac{(z_0 - \sqrt{\alpha_t} z_0)^2}{1 - \alpha_t} \right] \right\} \\ & \propto \exp \left\{ -\frac{1}{2} \left[\left(\frac{\alpha_t}{1 - \alpha_t} + \frac{1}{1 - \alpha_{t-1}} + \frac{1}{\tilde{\sigma}} \right) z_{t-1}^2 \right. \right. \\ & \quad \left. \left. + \left(\frac{-2\sqrt{\alpha_t} z_t}{1 - \alpha_t} + \frac{-2\sqrt{\alpha_{t-1}} z_0}{1 - \alpha_{t-1}} \right) z_{t-1} \right] + \Delta \right\} \end{aligned} \quad (31)$$

where Δ is a constant term.

Based on (31), the $\hat{\mu}$ of $p(z_{t-1}, z_t, z_0, c)$ is:

$$\hat{\mu} = -\frac{\frac{-2\sqrt{\alpha_t} z_t}{1 - \alpha_t} + \frac{-2\sqrt{\alpha_{t-1}} z_0}{1 - \alpha_{t-1}}}{2 \left(\frac{\alpha_t}{1 - \alpha_t} + \frac{1}{1 - \alpha_{t-1}} + \frac{1}{\tilde{\sigma}} \right)} \quad (32)$$

The above equation can be expanded in terms of z_t and z_0 as:

$$\begin{aligned} \hat{\mu} = & \frac{\sqrt{\alpha_t}(1 - \alpha_{t-1})\tilde{\sigma}}{\tilde{\sigma}(1 - \alpha_t) + (1 - \alpha_t)(1 - \alpha_{t-1})} z_t \\ & + \frac{(1 - \alpha_t)\tilde{\sigma}\sqrt{\alpha_{t-1}}}{\tilde{\sigma}(1 - \alpha_t) + (1 - \alpha_t)(1 - \alpha_{t-1})} z_0 \end{aligned} \quad (33)$$

Similar, the $\hat{\sigma}$ of $p(z_{t-1}, z_t, z_0, c)$ is:

$$\hat{\sigma} = \frac{\tilde{\sigma}(1 - \alpha_{t-1})}{\tilde{\sigma}(1 - \alpha_t) + (1 - \alpha_t)(1 - \alpha_{t-1})} (1 - \alpha_t) \quad (34)$$

Overall, the VLB can be written via (2) as:

$$\begin{aligned} \log p(x) & = \mathbb{E}_{q(z_{1:T} | z_0)} \log \frac{\prod_{t=1}^T p(z_t | z_{t-1}, c)}{q_\varphi(z_T) \prod_{t=1}^T q_\varphi(z_{t-1} | z_t, c)} \\ & = \mathbb{E} \left[-\log q_\varphi(z_T) + \sum_{t=2}^T \log \frac{p(z_t | z_{t-1}, c)}{q_\varphi(z_{t-1} | z_t, c)} \right. \\ & \quad \left. + \log \frac{p(z_1 | z_0, c)}{q_\varphi(z_0 | z_1, c)} \right] \end{aligned}$$

$$\begin{aligned}
&= \mathbb{E} \left[-\log q_\varphi(z_T) + \sum_{t=2}^T \log \frac{p(z_{t-1}|z_t, z_0, c)}{q_\varphi(z_{t-1}|z_t, c)} \right. \\
&\quad \left. + \sum_{t=2}^T \frac{p(z_t|z_0, c)}{p(z_{t-1}|z_0, c)} \right. \\
&\quad \left. + \log \frac{p(z_1|z_0, c)}{q_\varphi(z_0|z_1, c)} \right] \\
&= \mathbb{E} \left[-\log q_\varphi(z_T) + \sum_{t=2}^T \log \frac{p(z_{t-1}|z_t, z_0, c)}{q_\varphi(z_{t-1}|z_t, c)} \right. \\
&\quad \left. + \log \frac{p(z_T|z_0, c)}{p(z_1|z_0, c)} \right. \\
&\quad \left. + \log \frac{p(z_1|z_0, c)}{q_\varphi(z_0|z_1, c)} \right] \\
&= \mathbb{E} \left[\log \frac{p(z_T|z_0, c)}{q_\varphi(z_T)} + \sum_{t=2}^T \log \frac{p(z_{t-1}|z_t, z_0, c)}{q_\varphi(z_{t-1}|z_t, c)} \right. \\
&\quad \left. - \log q_\varphi(z_0|z_1, c) \right] \\
&\geq \mathcal{L}_{VLB} = \underbrace{\mathbb{E}_{q(z_1|z_0)} \log q_\varphi(z_0|z_1, c)}_{\mathcal{L}_1} \\
&\quad - \underbrace{\mathbb{E}_{q(z_{2:T}|z_0)} \text{DKL}(p(z_{t-1}|z_t, z_0, c) || q_\varphi(z_{t-1}|z_t, c))}_{\mathcal{L}_t}
\end{aligned} \tag{35}$$

where

$$\begin{aligned}
\mathcal{L}_t &= \mathbb{E}_{q(z_{2:T}|z_0)} \left[\frac{1}{2\tilde{\sigma}} \|\mu_\varphi(z_t, t, c) - \hat{\mu}\|^2 \right] \\
&= \frac{\tilde{\sigma}\bar{\alpha}_{t-1}}{2(1-\bar{\alpha}_{t-1})^2} \|\tilde{z}_\varphi(z_t, t, c) - z_0\|^2
\end{aligned} \tag{36}$$

which constrains $\tilde{z}_\varphi(z_t, t, c)$ to accurately predict z_0 .

As outlined in [14], [36], $\mathcal{L}_1$ is calculated as follows:

$$\mathcal{L}_1 = \|\tilde{z}_\varphi(z_1, 1, c) - z_0\|^2 \tag{37}$$

Remark: $\mathcal{L}_{VLB}$ (derived for FVD) yields a tighter variational lower bound than DDPM objective $\hat{\mathcal{L}}$ [14], i.e., $\mathcal{L}_{VLB} \geq \hat{\mathcal{L}}$.

Given

$$\begin{aligned}
\hat{\mathcal{L}} &= \|\tilde{z}_\varphi(z_1, 1) - z_0\|^2 - \left[\frac{1}{2} \left(\frac{\bar{\alpha}_{t-1}}{1-\bar{\alpha}_{t-1}} - \frac{\bar{\alpha}_t}{1-\bar{\alpha}_t} \right) \right. \\
&\quad \left. \times \|\tilde{z}_\varphi(z_t, t) - z_0\|^2 \right]
\end{aligned} \tag{38}$$

Assume that $\|\cdot\|$ is consistent with $\mathcal{L}_1$, and thus, a comparison of the coefficients between the two is required. The coefficient of $\hat{\mathcal{L}}$ is rewritten as:

$$\frac{\bar{\alpha}_{t-1}}{1-\bar{\alpha}_{t-1}} - \frac{\bar{\alpha}_t}{1-\bar{\alpha}_t} = \frac{\bar{\alpha}_{t-1}(1-\alpha_t)}{(1-\bar{\alpha}_{t-1})(1-\bar{\alpha}_t)} \tag{39}$$

The corresponding coefficient in $\mathcal{L}_t$ can be written as:

$$\frac{1}{1-\bar{\alpha}_{t-1}} \frac{(1-\alpha_t)\tilde{\sigma}\bar{\alpha}_{t-1}}{\tilde{\sigma}(1-\bar{\alpha}_t) + (1-\alpha_t)(1-\bar{\alpha}_{t-1})} \tag{40}$$

Algorithm 1: The Procedure of Integrating FVD into VAE-Based Methods.

Input: $G, \beta, \gamma, \alpha_{min}, \alpha_{max}$, and T .

Output: C .

1 **repeat**

2 **► Inference component**

3 Inference node representation z via Eqs. (8) ~ (10);

4 **► Cluster-driven conditional diffusion modules**

5 Noise is added to z in the forward process via Eq. (16);

6 Noise of z is removed in the reverse process via Eqs. (18) ~ (20);

7 **► Graph wavelet transform**

8 Perform secondary signal modulation on z to obtain $\hat{z}$ via Eqs. (22) ~ (23);

9 **► Generation component**

10 Reconstruct origin graph $\hat{\mathbf{A}}$ via Eq. (24) ;

11 **► Optimization**

12 Calculate $\mathcal{L}$ via Eq. (25) and Eqs. (35) ~ (43);

13 **until** Converge;

14 **return** C .

Since $0 < \alpha_t < 1$ and $\bar{\alpha}_{t-1} > \bar{\alpha}_t$ (equivalently $\alpha_1 > \alpha_2 > \dots > \alpha_T > 0$ with $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$), we have $(1-\alpha_t)(1-\bar{\alpha}_{t-1}) > 0$. Hence

$$\tilde{\sigma}(1-\bar{\alpha}_t) < \tilde{\sigma}(1-\bar{\alpha}_t) + (1-\alpha_t)(1-\bar{\alpha}_{t-1}) \tag{41}$$

which implies

$$\frac{\tilde{\sigma}}{\tilde{\sigma}(1-\bar{\alpha}_t) + (1-\alpha_t)(1-\bar{\alpha}_{t-1})} \leq \frac{\tilde{\sigma}}{\tilde{\sigma}(1-\bar{\alpha}_t)} = \frac{1}{1-\bar{\alpha}_t} \tag{42}$$

(39) ~ (42) together imply that the per-step coefficient of $\mathcal{L}_{VLB}$ is no larger than that of $\hat{\mathcal{L}}$. Since both objectives subtract these terms under the same norm assume, it follows that $\mathcal{L}_{VLB} \geq \hat{\mathcal{L}}$.

Total optimization function: For VAEs integrated with FVD, the overall optimization objective comprises two components: the optimization objective of the integrated method (including ELBO) and the optimization objective of FVD (i.e., VLB):

$$\mathcal{L} = \mathcal{L}_{ELBO} + \gamma \mathcal{L}_{VLB} \tag{43}$$

where γ is a hyperparameter used to control the contribution of different optimization objectives.

Finally, clusters are derived based on the partitioning pattern of the integrated methods. For methods specifically designed for clustering tasks, cluster structures are identified in an end-to-end manner. In contrast, for general-purpose methods not explicitly tailored for clustering, K-means is typically employed to partition cluster structures. We outline the complete procedure of integrating FVD into existing VAE-based methods in Algorithm 1.

V. EXPERIMENTS

A. Experiment Setup

Datasets: We use Cora, Citeseer, Pubmed, ACM and Blog-catalog as benchmark datasets to conduct our experiments, and the detailed data are shown in Table II.

Baselines: FVD is designed to address the limitations of integrating DDPM into VAE-based methods. To this end, we

TABLE II
THE STATISTIC OF DATASETS. $Avg.C$ DENOTES THE AVERAGE CLUSTERING COEFFICIENT.

Datasets	N	# Edge	M	k	$Avg.C$
Cora	2,708	5,278	1,433	7	0.2406
Citeseer	3,327	4,552	3,703	6	0.1414
Pubmed	19,717	44,324	500	3	0.060
ACM	3,025	26,256	1,870	3	0.5491
Blogcatalog	5,196	171,743	8,189	6	0.1223

integrate FVD into a variety of representative and state-of-the-art VAE-based methods to conduct node clustering experiments. These methods include methods specifically designed for node clustering tasks, such as VGAECD [12], GM-VGAE [37], CV-GAE [11] and VGAER [7], as well as general-purpose methods not explicitly tailored for clustering, including VGAE [6], VGNAE [38], DGVAE [39] and RWR-VGAE [40]. Detailed descriptions of these methods are provided in Section VI.

Parameter setting and evaluation metrics: FVD is a plug-and-play method, meaning it can be seamlessly integrated into any VAE-based approach without requiring modifications to the original module. Therefore, all baseline methods either adopt the parameter settings described in their papers or, if they use different datasets, fine-tune them based on existing settings. The parameter settings for FVD need to focus on two modules. For the cluster-driven conditional diffusion module, the diffusion step T is optimized within the range [50, 100, 300, 500], with $T = 100$ being effective for most scenarios. To improve generalization, we apply a dropout layer during the reverse process of FVD, with the dropout ratio searched within the range $\{0.2, 0.3, 0.4, 0.5\}$. For γ in the total optimization function, the optimal value is determined by searching within the range $\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$. For the graph wavelet transform module, we fix the parameter K to 2, and set θ to [1, -0.5] with reference to [31]. Detailed parameter settings used in our experiments are provided in Appendix A, available online. All experiments are performed on a PC running 64-bit Windows 10, equipped with a 3.40 GHz Intel Core i7-6700 CPU and 64 GB of RAM. Each method is run three times (with the random seed fixed during the first run), and the mean and standard deviation of the performance are reported in Section 2. For the remaining experimental sections, results are reported using a fixed random seed. Additionally, the node clustering performance of all methods is evaluated using four widely adopted metrics: ACC, NMI, ARI and F1.

B. Node Clustering Performance

We apply FVD to various VAE-based methods, and the node clustering results are summarized in Table III. We have the following observations and analysis:

(1) *Comprehensive experiments reveal that FVD effectively enhances the expressiveness of the latent space in VAEs, driving improved clustering performance:* The performance comparison of all VAE-based methods before and after integrating FVD shows that FVD enables these methods to achieve more competitive results in node clustering tasks. On an overall scale, methods integrated with FVD achieve average performance improvements of 2.9, 2.15, 3.19 and 2.76 points in ACC, NMI,

ARI and F1, respectively. In some cases, the improvements are even more pronounced, reaching 5.18, 4.16, 7.32 and 5.09 points for the same metrics. In addition to these improvements, the integration of FVD further improves the robustness of the methods, as demonstrated by significantly lower standard deviations across a majority of performance metrics. For example, after integrating FVD, VGAE experiences approximately 2, 3 and 3 points reductions in standard deviation for ACC, NMI and ARI, respectively. Similarly, CVGAE maintains stable performance post-integration, with no observable fluctuations in its overall results. These findings suggest that FVD not only boosts overall performance but also enhances stability in clustering tasks. In general, the experiments demonstrate that the combination of graph wavelet transform and cluster-driven conditional diffusion substantially enhances the performance of VAE-based methods.

(2) *FVD demonstrates a stronger affinity for methods explicitly designed to address clustering tasks:* We analyze the impact of integrating FVD into methods specifically tailored for clustering tasks, and the results highlight significant enhancements in their ability to capture cluster structures. Specifically, on an overall scale, methods combined with FVD achieve average performance improvements of 3.85, 2.4, 3.73 and 3.4 points in ACC, NMI, ARI and F1, respectively. Notably, GM-VGAE exhibits remarkable gains on the ACM dataset, with improvements of 19.6, 13.7, 25.7 and 18.4 points across the same metrics. Similarly, VGAECD benefits significantly from FVD integration, achieving competitive boosts of 8.4, 7.3, 12.5 and 7.9 points. The affinity of FVD for clustering-oriented methods can be attributed to the complementary strengths of the Student's t-distribution and the Gaussian mixture distribution. Unlike the Gaussian distribution, the Student's t-distribution demonstrates heightened sensitivity to cluster boundary nodes, enabling it to capture critical boundary information that Gaussian mixture models often overlook. During the reverse process in FVD, this sensitivity facilitates the recovery of previously neglected cluster boundary details while concurrently preserving fine-grained node information.

(3) *FVD shows flexibility across a variety of common scenarios:* Our experiments also integrate FVD into methods not originally designed for node clustering. Although the degree of improvement is smaller compared to methods explicitly designed for clustering, these methods still benefit from the signal modulation capabilities of FVD. Specifically, after two rounds of signal modulation and the incorporation of cluster information, these methods achieve average performance improvements of 1.95, 1.89, 2.65, and 2.12 points in ACC, NMI, ARI and F1, respectively. On specific datasets, the improvements are even more pronounced, with competitive gains of 3.68, 3.15, 4.15 and 3.9 points across these metrics. We attribute these results to the discovery of the LGH pattern and the design of the guided VLB. The noise addition and removal processes in the diffusion module can be interpreted as frequency signal processing, where aligning the signal modulation results with the LGH pattern allows existing methods to benefit from frequency-adjusted node representations. Moreover, the cluster-driven VLB enhances the expressiveness of the latent space, enabling it to encode cluster information more effectively. This process implicitly embeds

TABLE III
NODE CLUSTERING RESULTS (%): MEAN $\pm$ STANDARD DEVIATION (BOLD: BEST). THE ABBREVIATION FOR RWR-VGAE IS RWR-,
AND FOR GM-VGAE, IT IS GM-.

Datasets	Metrics	VGAE +FVD	VGNAE +FVD	VGAECD +FVD	DGVAE +FVD	CVGAE +FVD	RWR-VGAE +FVD	VGAER +FVD	GM-VGAE +FVD
Cora	ACC	65.0 $\pm$ 1 67.1$\pm$2	73.6 $\pm$ 0 74.0$\pm$1	67.1 $\pm$ 4 69.2$\pm$3	67.6 $\pm$ 3 68.8$\pm$3	75.1 $\pm$ 0 75.0 $\pm$ 0	61.8 $\pm$ 2 64.1$\pm$5	63.9 $\pm$ 0 66.5$\pm$1	62.5 $\pm$ 1 64.8$\pm$1
	NMI	47.4 $\pm$ 1 49.4$\pm$1	55.1 $\pm$ 0 57.0$\pm$1	52.2 $\pm$ 2 53.0$\pm$1	51.0 $\pm$ 2 52.8$\pm$2	57.0 $\pm$ 0 57.2$\pm$0	44.7 $\pm$ 1 46.1$\pm$1	45.3 $\pm$ 0 45.5$\pm$0	45.3 $\pm$ 2 48.2$\pm$1
	ARI	40.3 $\pm$ 1 43.5$\pm$3	52.5 $\pm$ 0 54.0$\pm$2	45.6 $\pm$ 4 46.2$\pm$2	45.1 $\pm$ 0 45.7$\pm$4	55.2 $\pm$ 0 55.9$\pm$0	37.3 $\pm$ 2 38.8$\pm$3	36.9 $\pm$ 0 39.9$\pm$3	38.1 $\pm$ 0 42.3$\pm$1
	F1	64.2 $\pm$ 0 65.4$\pm$2	71.8 $\pm$ 0 71.9$\pm$1	65.7 $\pm$ 4 64.4 $\pm$ 3	65.6 $\pm$ 4 66.7$\pm$4	73.0 $\pm$ 0 72.2 $\pm$ 0	59.9 $\pm$ 2 62.5$\pm$5	64.3 $\pm$ 0 65.5$\pm$1	60.9 $\pm$ 1 61.6$\pm$2
Citeseer	ACC	49.1 $\pm$ 3 52.5$\pm$0	65.5 $\pm$ 1 67.4$\pm$0	53.5 $\pm$ 7 58.1$\pm$8	52.2 $\pm$ 2 56.5$\pm$2	66.1 $\pm$ 0 68.0$\pm$0	48.0 $\pm$ 2 53.1$\pm$0	40.1 $\pm$ 0 41.8$\pm$0	43.7 $\pm$ 3 47.3$\pm$1
	NMI	24.2 $\pm$ 2 25.2$\pm$1	39.2 $\pm$ 0 41.4$\pm$0	30.6 $\pm$ 4 32.7$\pm$4	28.0 $\pm$ 0 30.1$\pm$1	40.1 $\pm$ 0 40.7$\pm$0	24.0 $\pm$ 1 26.4$\pm$0	21.5 $\pm$ 0 23.1$\pm$1	19.8 $\pm$ 1 24.2$\pm$0
	ARI	21.1 $\pm$ 2 23.0$\pm$0	40.0 $\pm$ 0 42.7$\pm$0	27.0 $\pm$ 7 30.7$\pm$5	21.6 $\pm$ 3 29.7$\pm$2	41.2 $\pm$ 0 41.5$\pm$0	20.9 $\pm$ 0 24.8$\pm$0	10.7 $\pm$ 0 10.5$\pm$0	16.6 $\pm$ 1 18.8$\pm$1
	F1	46.0 $\pm$ 2 50.9$\pm$0	61.4 $\pm$ 1 62.8$\pm$0	50.6 $\pm$ 8 52.1$\pm$6	50.5 $\pm$ 0 53.4$\pm$2	62.7 $\pm$ 0 62.5 $\pm$ 0	44.9 $\pm$ 3 51.3$\pm$0	37.8 $\pm$ 0 40.2$\pm$0	42.2 $\pm$ 3 47.2$\pm$1
Pubmed	ACC	63.5 $\pm$ 1 65.0$\pm$1	64.1 $\pm$ 0 64.6$\pm$0	57.1 $\pm$ 6 60.2$\pm$4	68.6 $\pm$ 1 69.6$\pm$0	66.3 $\pm$ 0 68.8$\pm$0	63.9 $\pm$ 1 65.9$\pm$1	65.0 $\pm$ 0 66.1$\pm$0	65.7 $\pm$ 2 66.2$\pm$0
	NMI	22.1 $\pm$ 3 23.7$\pm$1	21.9 $\pm$ 0 22.1$\pm$0	20.5 $\pm$ 6 21.8$\pm$3	29.2 $\pm$ 1 30.8$\pm$0	29.9 $\pm$ 0 30.1$\pm$0	24.0 $\pm$ 1 24.7$\pm$1	29.9 $\pm$ 0 29.8$\pm$0	24.1 $\pm$ 2 26.0$\pm$2
	ARI	21.5 $\pm$ 2 23.8$\pm$1	23.0 $\pm$ 0 23.7$\pm$0	20.0 $\pm$ 6 21.8$\pm$2	29.9 $\pm$ 2 32.2$\pm$0	28.2 $\pm$ 0 30.6$\pm$0	22.7 $\pm$ 1 24.1$\pm$0	27.9 $\pm$ 0 29.3$\pm$0	24.8 $\pm$ 3 25.9$\pm$2
	F1	63.6 $\pm$ 1 65.3$\pm$1	63.0 $\pm$ 0 63.6$\pm$0	52.8 $\pm$ 11 57.2$\pm$9	68.2 $\pm$ 0 68.3$\pm$0	65.8 $\pm$ 0 68.1$\pm$0	64.4 $\pm$ 1 65.9$\pm$0	64.0 $\pm$ 0 64.9$\pm$0	65.7 $\pm$ 2 65.9$\pm$0
ACM	ACC	84.0 $\pm$ 0 85.9$\pm$0	89.8 $\pm$ 0 90.5$\pm$0	65.1 $\pm$ 2 73.5$\pm$7	84.5 $\pm$ 1 85.1$\pm$1	86.0 $\pm$ 0 86.0$\pm$0	84.8 $\pm$ 1 85.7$\pm$1	47.2 $\pm$ 0 56.6$\pm$0	62.0 $\pm$ 1 81.6$\pm$1
	NMI	53.0 $\pm$ 1 55.7$\pm$2	66.6 $\pm$ 0 67.9$\pm$0	33.9 $\pm$ 1 41.2$\pm$7	53.5 $\pm$ 3 54.5$\pm$2	55.8 $\pm$ 0 56.1$\pm$1	55.2 $\pm$ 0 56.5$\pm$1	17.7 $\pm$ 0 23.4$\pm$1	33.4 $\pm$ 0 47.1$\pm$3
	ARI	57.7 $\pm$ 2 62.4$\pm$1	72.2 $\pm$ 1 73.8$\pm$0	27.0 $\pm$ 0 39.5$\pm$10	59.0 $\pm$ 4 60.6$\pm$3	62.8 $\pm$ 0 62.6 $\pm$ 1	60.1 $\pm$ 2 63.1$\pm$1	0.5 $\pm$ 0 10.2$\pm$0	27.1 $\pm$ 1 52.8$\pm$3
	F1	84.0 $\pm$ 0 85.9$\pm$0	89.8 $\pm$ 0 90.4$\pm$0	66.3 $\pm$ 2 74.2$\pm$6	84.5 $\pm$ 1 85.1$\pm$1	86.0 $\pm$ 0 85.9 $\pm$ 0	84.8 $\pm$ 1 85.7$\pm$1	43.9 $\pm$ 1 54.4$\pm$0	63.2 $\pm$ 1 81.6$\pm$1
Blogcatalog	ACC	41.0 $\pm$ 1 44.5$\pm$1	53.2 $\pm$ 2 52.3$\pm$0	37.8 $\pm$ 0 40.6$\pm$0	40.7 $\pm$ 4 44.3$\pm$1	41.0 $\pm$ 2 43.1$\pm$0	40.2 $\pm$ 2 43.2$\pm$0	38.0 $\pm$ 1 43.2$\pm$4	34.0 $\pm$ 4 37.6$\pm$4
	NMI	25.4 $\pm$ 1 28.5$\pm$0	32.2 $\pm$ 2 33.3$\pm$2	22.1 $\pm$ 1 23.9$\pm$1	26.1 $\pm$ 3 28.8$\pm$1	23.0 $\pm$ 0 23.1$\pm$0	23.3 $\pm$ 3 29.0$\pm$1	24.2 $\pm$ 0 23.9$\pm$0	17.1 $\pm$ 3 20.4$\pm$1
	ARI	16.6 $\pm$ 1 19.5$\pm$0	24.4 $\pm$ 2 25.8$\pm$2	13.3 $\pm$ 0 15.0$\pm$1	16.2 $\pm$ 3 19.2$\pm$0	13.4 $\pm$ 1 14.5$\pm$0	13.6 $\pm$ 4 18.3$\pm$0	15.8 $\pm$ 0 16.3$\pm$0	8.6 $\pm$ 2 11.0$\pm$0
	F1	37.9 $\pm$ 1 44.6$\pm$1	51.4 $\pm$ 4 49.3$\pm$1	37.5 $\pm$ 1 38.6$\pm$1	39.9 $\pm$ 6 43.2$\pm$2	38.1 $\pm$ 3 40.8$\pm$0	38.0 $\pm$ 2 44.0$\pm$0	36.9 $\pm$ 1 42.8$\pm$4	32.2 $\pm$ 3 37.4$\pm$4

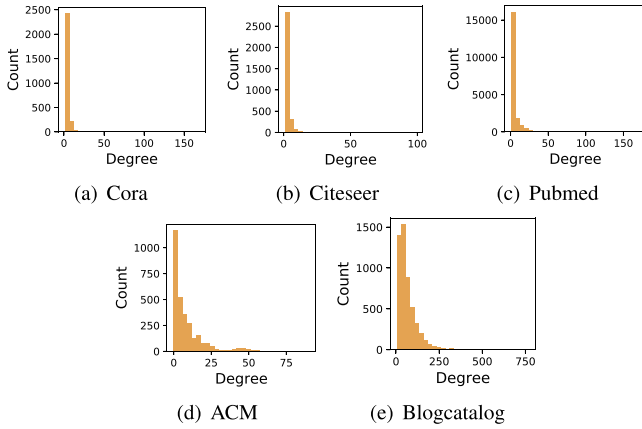


Fig. 6. Visualization of the degree distributions for all datasets.

cluster structures into the node representations, further boosting their performance in node clustering tasks.

(4) *FVD performs better on graphs with strong cluster patterns and broad degree distributions*: To further explore the conditions under which FVD is most effective, we analyze the relationship between the structural properties of the graph and the performance gains obtained by integrating FVD. Fig. 6 and Table II respectively present the degree distributions and average clustering coefficients of all datasets. We observed that the most significant performance improvements occur on the ACM, while the gains on Pubmed are comparatively limited. By examining the degree distributions and average clustering coefficients, we conclude a consistent pattern: FVD tends to yield more substantial improvements on datasets with higher average clustering coefficients and broader degree distributions. In contrast, its benefits are less pronounced on datasets with lower clustering coefficients and more narrow degree structures. A possible explanation is that the presence of clear local structural patterns and well-defined cluster boundaries allows FVD to effectively leverage this information to guide the reverse diffusion

process. In this case, the Student's t-distribution facilitates the preservation of local variations and mitigates cluster collapse.

C. Ablation Experiments and Motivation Analysis

We conduct ablation experiments to explore and validate the two motivations behind FVD.

1) *Cluster-driven conditional diffusion module facilitates the latent space in capturing and reflecting cluster structure information*: To validate this motivation, we expand the experiments presented in the Section III-C. Specifically, we measure intra-cluster compactness and inter-cluster separation after introducing the cluster-driven conditional diffusion module. The results, summarized in Table V, show significant improvements in both metrics compared to the initial distances, confirming ability of the module to enhance cluster cohesion and separation. Furthermore, a comparison with the DDPM results in Table I indicates that our conditional diffusion module offers a more effective mechanism for the latent space to reflect cluster structure information.

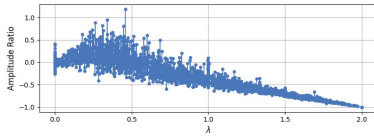
Table IV compares the performance of various baseline methods with and without the cluster-driven conditional diffusion module (i.e., Wavelet: $\times$ and Conditional: $\checkmark/\times$). The results reveal that, in most cases, incorporating the conditional diffusion module leads to noticeable performance gains. For methods not specifically designed for clustering tasks, these improvements are sometimes significant. For example, VGAE on the Citeseer dataset shows remarkable increases in ACC, NMI, ARI and F1 by 7.12, 2.75, 6.76 and 5.32 points, respectively. Even for clustering-specific methods, such as VGAECD on the same dataset, the module provides consistent benefits, with improvements of 1.9, 0.97, 1.31 and 0.52 points in ACC, NMI, ARI and F1, respectively. These findings underscore the versatility and adaptability of the cluster-driven conditional diffusion module across different types of methods. They also validate the rationality and effectiveness of our first motivation.

2) *Graph wavelet transform, as a secondary signal modulation, complements the diffusion model in enhancing*

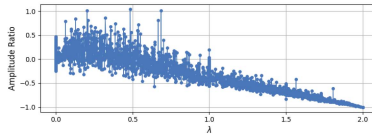
TABLE IV

ABLATION EXPERIMENTS. ALL PERFORMANCE RESULTS (%) ARE REPORTED USING THE SAME RANDOM NUMBER SEED. WHEN BOTH THE CLUSTER-DRIVEN **CONDITIONAL** DIFFUSION MODULE AND THE GRAPH **WAVELET** TRANSFORM ARE INCLUDED, THE CONFIGURATION CORRESPONDS TO OUR PROPOSED FVD METHOD. CONVERSELY, WHEN NEITHER IS INCLUDED, IT CORRESPONDS TO THE VANILLA DDPM.

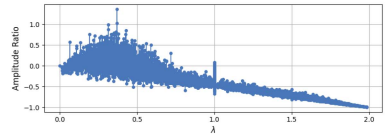
Conditional	Wavelet	Cora				Citeseer				Pubmed				ACM				Blogcatalog			
		ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1
VGAE																					
✓	✓	67.21	49.54	41.86	66.55	51.85	25.02	22.24	51.02	64.97	22.51	23.44	64.68	84.99	53.42	60.34	85.04	45.28	29.36	19.95	43.89
✓	×	59.31	44.23	36.23	58.24	54.01	25.49	25.17	51.31	64.09	21.92	22.36	64.37	80.4	43.87	50.1	80.44	47.92	28.86	19.54	47.4
×	×	54.8	45.33	34.71	52.91	46.89	22.74	18.41	45.99	59.51	17.28	16.05	59.73	74.48	40.48	41.57	74.57	39.41	25.05	14.63	38.99
VGNAE																					
✓	✓	75.15	58.58	55.98	72.93	67.36	41.31	42.75	62.75	64.49	22.15	23.88	64.17	90.74	68.55	74.52	90.71	52.91	35.16	28.05	49.26
✓	×	73.52	56.03	53.23	71.34	66.61	40.62	41.61	62.57	66.49	24.43	26.29	65.46	90.78	68.85	74.67	90.76	52.73	35.7	28.38	49.09
×	×	72.56	52.25	49.74	70.93	67.06	39.09	40.86	63.23	63.89	21.07	21.99	63.54	89.62	65.49	71.57	89.57	53.16	32.87	26.41	50.91
VGAECD																					
✓	✓	72.49	54.75	48.98	68.34	62.88	35.93	34.38	57.01	55.18	25.39	24.81	46.89	68.99	36.71	33.06	69.81	41.19	23.45	14.01	37.12
✓	×	72.42	54.67	48.58	68.19	62.28	34.62	33.21	56.37	55.01	24.88	24.73	45.83	64.43	31.84	26	65.72	40.15	22.59	13.28	36.55
×	×	63.22	48.56	43.75	61.12	60.38	33.65	31.9	55.85	53.18	21.93	21.98	42.84	68.2	32.68	27.85	66.58	35.95	20.94	12.7	35.61
DGVAE																					
✓	✓	72.71	54.43	49.4	70.7	54.07	28.43	27.93	51.37	69.61	30.65	32.6	68.2	85.45	54.09	61.05	85.51	46.19	27.04	19.08	45.54
✓	×	63.48	48.72	41.03	63.72	47.16	25.96	22.94	43.96	69.06	30	30.77	67.81	82.88	49.65	55.54	82.84	45.03	28.92	19.63	42
×	×	67.47	51.57	46.52	65.5	53.68	27.21	24.93	51.86	68.49	29.53	29.81	67.39	83.27	51.72	56.7	83.25	43.4	26.91	17.09	40.45
CVGAE																					
✓	✓	75.37	57.32	56.29	72.52	67.63	40.26	40.83	62.59	68.89	30.27	30.5	68.31	86.51	57.75	63.97	86.41	43.28	23.32	14.58	40.81
✓	×	75.18	56.89	56.15	72.18	66.04	37.28	38.65	60.4	66.9	27.76	28.07	66.2	83.37	53.79	56.16	83.32	42.63	23.61	13.86	41.41
×	×	74.82	55.48	55.66	72.22	61.68	32.29	30.81	54.38	66.3	26.84	27.22	65.63	85.02	54.66	60.35	85.01	42.74	22.38	13.03	40.15
RWR-VGAE																					
✓	✓	62.52	46.42	38.62	58.87	52.75	25.89	24.88	50.66	65.51	24.99	24.81	65.84	86.94	58.59	65.18	86.94	42.59	28.41	17.33	43.25
✓	×	61.37	48.79	40.81	61.2	55.46	28.12	25.86	53.65	63.79	24.07	22.73	64.22	84.86	54.21	60.18	84.95	46.65	27.77	17.26	45.94
×	×	61.04	45.27	40.69	57.97	48.45	25.14	21.02	47.37	63.17	19.82	20.73	62.8	85.19	54.38	60.91	85.26	43.67	27.19	18.66	43.73
VGAE-R																					
✓	✓	68.46	45.81	43.27	66.66	42.08	21.6	11.43	40.28	66.67	30.14	30.2	65.34	56.43	22.25	10.12	54.48	41.19	24.15	15.58	40.91
✓	×	59.9	43.05	37.4	58.72	41.84	17.89	13.36	41.02	67.12	29.22	30.1	65.85	59.67	25.86	13.9	58.43	37.74	24.52	15.82	37.57
×	×	48.23	34.4	25.62	46.37	38.89	15.53	12.07	37.57	67.05	29.37	29.85	65.63	56.07	20.75	10.18	54.79	42.13	24.03	16.46	41.5
GM-VGAE																					
✓	✓	64.14	50.32	43.93	59.6	48.54	23.64	19.55	48.05	65.43	23.49	24.17	64.98	79.6	43.24	48.57	79.53	35.22	19.73	11.36	34.61
✓	×	58.38	39.89	32.79	56.97	47.01	22.34	19.73	44.71	63.1	23.46	21.36	63.15	64.3	33.41	30.79	65.06	34.95	17.83	9.97	35.35
×	×	63.18	47.21	38.84	61.54	39.98	17.9	11.74	38.9	64.98	22.21	23.81	64.2	60.6	32.24	25.29	62.15	37.14	19.53	10.29	38.27



(a) Cora



(b) Citeseer



(c) Pubmed

Fig. 7. Amplification ratio of signal amplitude before and after graph wavelet transform modulation. After applying graph wavelet transform, 18.17%, 23.08% and 8.44% of the nodes in Cora, Citeseer, and Pubmed, respectively, exhibit increased signal amplitudes.

TABLE V

EVALUATING THE INFLUENCE OF CLUSTER-DRIVEN **CONDITIONAL** DIFFUSION ON INTRA-CLUSTER COMPACTNESS AND INTER-CLUSTER SEPARATION. BOTH ▼ AND ▲ REPRESENT POSITIVE RESULTS.

Methods	node2vec	Conditional	node2vec	Conditional
Metrics	Intra-cluster distance		Inter-cluster distance	
Cora	41.55	35.13▼	8.52	10.94▲
Citeseer	43.94	42.41▼	9.31	12.40▲
Pubmed	63.07	61.96▼	28.17	34.05▲

the latent space expressiveness: To validate this motivation, we extend the experiments in Section III-B by analyzing the impact of the graph wavelet transform on signal amplitudes. The changes are visualized in Fig. 7, which shows that while vanilla DDPM amplifies signals across the frequency spectrum, the graph wavelet transform suppresses high-frequency components while maintaining meaningful amplification in the low-frequency range. Specifically, after applying the graph wavelet transform, only 18.17%, 23.08% and 8.44% of the nodes in Cora, Citeseer and Pubmed, respectively, exhibit increased signal amplitude, which is primarily concentrated in the low-frequency range. Importantly, the amplitude variations observed before and after applying this transform align with the LGH pattern, underscoring the efficacy of the graph wavelet transform in enhancing signal processing.

The ablation experiments summarized in Table IV compare the performance of various baseline methods before and after applying the graph wavelet transform (i.e., Conditional: ✓ and Wavelet: ✓/×). The results indicate that baseline methods integrated with FVD generally demonstrate competitive performance improvements. For example, on the Cora dataset, VGAE achieves notable gains of 7.9, 5.31, 5.63 and 8.31 points in ACC, NMI, ARI and F1, respectively. Similarly, GM-VGAE, which is specifically designed for clustering tasks, shows significant enhancements on the same dataset, with increases of 5.76, 10.43, 11.14 and 2.63 points in these metrics. These findings highlight that the graph wavelet transform effectively amplifies low-frequency signals while suppressing high-frequency components, aligning with spectral characteristics of VAEs. However, in certain cases, relying solely on the conditional diffusion module still achieves considerable performance gains. This variation can be attributed to differences in how individual methods affect signal amplitudes, potentially leading to blurred clustering boundaries. Specifically, methods that inherently suppress high-frequency signals or struggle to capture cluster boundaries may experience diminished benefits from the wavelet transform, as it could inadvertently lead to the loss of critical boundary information. Overall, although the full integration of FVD may not always produce optimal results, methods enhanced with this

plug-in consistently achieve competitive performance compared to their original implementations.

D. Effectiveness Analysis Between FVD and DDPM

Table IV also shows the performance improvements achieved by FVD and vanilla DDPM across various VAE-based methods. Specifically, Conditional: $\checkmark$ and Wavelet: $\checkmark$ represent the configuration using FVD, whereas Conditional: $\times$ and Wavelet: $\times$ correspond to the vanilla DDPM setup.

We observe that integrating FVD yields more substantial performance improvements compared to DDPM integration. On average, ACC, NMI, F1 and ARI improve by 3.61, 3.53, 4.28 and 3.4 points, respectively. For certain methods, these enhancements are even more pronounced, with gains of 20.23, 12.94, 23.28 and 20.29 points across the four metrics. The competitive performance of FVD can be attributed to its cluster-driven conditional diffusion, coupled with a frequency modulation strategy aligned with the LGH pattern. Unlike DDPM, which primarily focuses on refining node representations, FVD is specifically designed to meet the unique demands of node clustering by effectively capturing and enhancing cluster-specific structural information. However, in some cases, FVD achieves near-optimal performance improvements or, occasionally, exhibits minor drawbacks. These outcomes likely arise from the design sensitivities of certain models to signals in specific frequency bands. For instance, VGAE replaces node attributes with a modularity matrix for inference and generation. From a graph signal processing perspective, this approach results in lower signal amplitudes near $\lambda = 1$, because node attribute information is excluded, leaving only topology data. While topology data reflects the overall graph structure, it is less effective at delineating cluster boundaries. Consequently, the secondary signal modulation introduced by FVD may inadvertently diminish the clarity of these boundaries, thereby increasing the risk of cluster collapse.

E. Hyperparameter Analysis of γ in Joint Training

We recognize that integrating diffusion model into variational autoencoder frameworks often requires some parameter tuning [14], particularly to balance the contributions of VLB and ELBO. This challenge arises from the inherent differences and interdependence of their optimization objectives. Specifically, the diffusion model focuses on capturing fine-grained information from the node representations output by the variational autoencoder, while the variational autoencoder reconstructing the original graph structure as accurately as possible through the new node representations output by diffusion model [41]. Additionally, the slower convergence of diffusion model compared to variational autoencoders further complicates the balancing process [42], [43]. Without careful parameter adjustment, the training process may become dominated by one model, leading to suboptimal overall performance. To investigate this issue, we use VGAE as a representative example and conduct experiments on the Cora, Citeseer, Pubmed, ACM and Blogcatalog. Specifically, we expand the range of γ to explore its impact, evaluating values $\{10^0, 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$. The

results, visualized in Fig. 8, reveal that without a balancing parameter between VLB and ELBO, VGAE fails to achieve effective training. Furthermore, overly small γ may hinder the training of FVD on complex datasets (e.g., Pubmed), degrading the overall performance of VGAE. Analyzing Fig. 8 alongside the parameter settings in Appendix A, available online, we find that $\gamma \in \{10^{-3}, 10^{-4}\}$ generally works well for most methods across diverse datasets. For methods requiring pre-training, multiple attempts to adjust γ may be necessary to achieve optimal performance. Nevertheless, the choice of γ is relatively straightforward, as suitable values are typically confined to the range $\{10^{-2}, 10^{-3}, 10^{-4}\}$.

F. Dropout Rate Analysis in Reverse Process

During the reverse process of FVD, noise is gradually removed to inference the original samples. In this process, the dropout layer serves as a regularization mechanism, ensuring robustness in the cluster task. To determine the suitable range of dropout rates, we use VGAE as a case study and analyze its performance across multiple datasets. The experimental results, shown in Fig. 9, evaluate VGAE with dropout rates ranging from 0% to 50%. While the default dropout rate for VGAE (as specified in Appendix A), available online is set to 0%, our observations reveal that optimal performance is often achieved within the range of 5% to 25%, as illustrated in Fig. 9(b) and (e). Notably, excessively high dropout rates suppress critical structural information, leading to ineffective method training, as shown in Fig. 9(c). These findings suggest that smaller dropout rates are sufficient for most scenarios. However, for methods requiring pre-training, careful tuning the dropout rate is necessary.

G. Analysis of the Trade-Off Between Efficiency and Performance

The time complexity of FVD primarily consists of two components: the cluster-driven conditional diffusion module and the graph wavelet transform. The diffusion module has a complexity of $\mathcal{O}(TNM)$. The graph wavelet transform, when implemented using Chebyshev polynomial approximation, has a linear complexity of $\mathcal{O}(KE)$, where E is the number of edges. To more intuitively demonstrate the runtime efficiency of FVD, we compare the training time of the vanilla VGAE and VGAE integrated with FVD over 100 epochs. The results are shown in Fig. 10(a). Across all datasets, the time cost of VGAE+FVD is only slightly higher than that of the original VGAE, indicating that FVD enhances performance while keeping computational costs manageable. A key reason for the low overhead is our use of Chebyshev approximation to compute the graph wavelet transform, which avoids costly eigenvalue decomposition and wavelet basis construction.

To further compare the time overhead and performance benefits of FVD with full graph wavelet transform (denoted as FVD_{Eigen.}) versus the standard Chebyshev-approximated FVD, we conduct additional experiments and statistical analyses. As shown in Fig. 10(b), we first compare the preprocessing time required by the two implementations across different datasets.

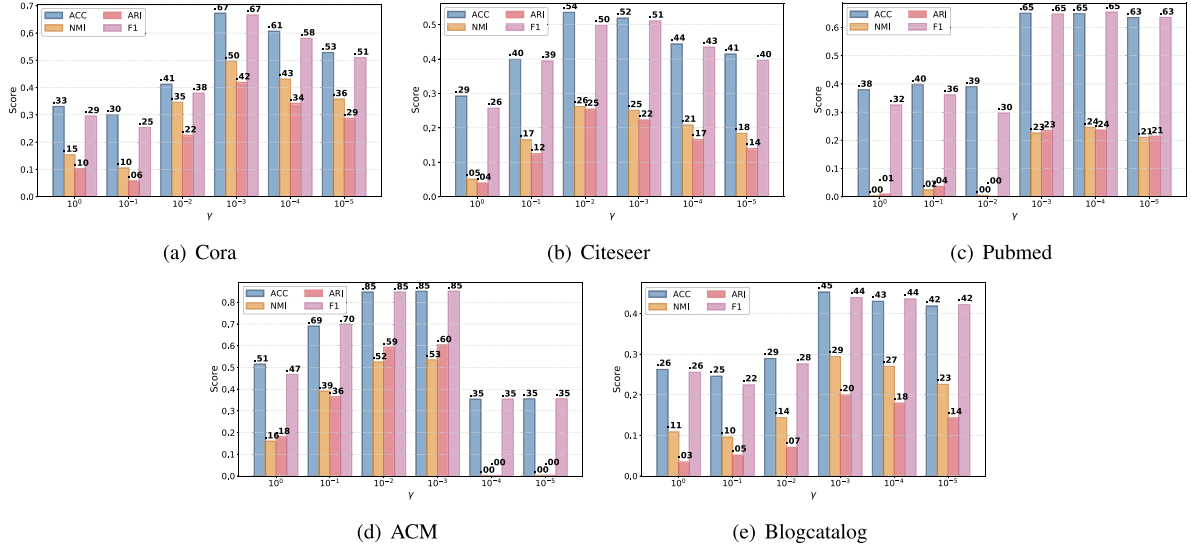


Fig. 8. Exploring the influence of γ on VGAE-FVD joint training performance.

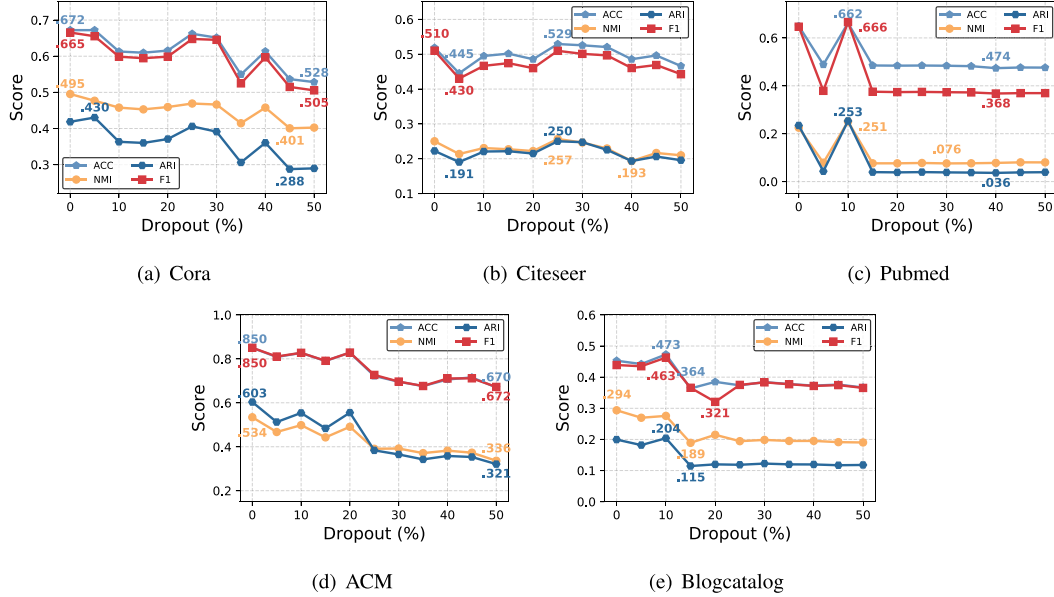


Fig. 9. Impact of different dropout rates on VGAE performance, where we set dropout rates $\in \{0\%, 5\%, 10\%, 15\%, 20\%, 25\%, 30\%, 35\%, 40\%, 45\%, 50\%\}$.

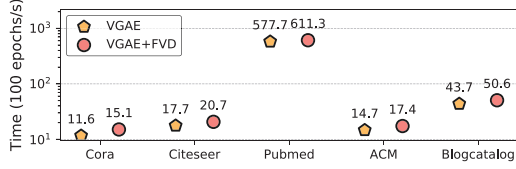
It can be observed that on large-scale datasets such as Pubmed, the computational cost of eigendecomposition increases significantly, which makes FVD_{Eigen} not suitable for large graphs. Next, we evaluate the performance improvements brought by FVD_{Eigen} and standard FVD when integrated into VGAE, as summarized in Table VI. While FVD_{Eigen} generally achieves slightly higher performance gains, the additional time overhead becomes non-negligible, particularly on large graphs. Considering that the standard FVD achieves competitive improvements with minimal computational cost, we recommend it for achieving a practical trade-off between efficiency and effectiveness.

Furthermore, FVD can be viewed as signal modulators, expanding the toolbox of graph signal processing. Existing graph signal processing methods typically fall into two categories: frequency response function design [44], [45], [46] and graph

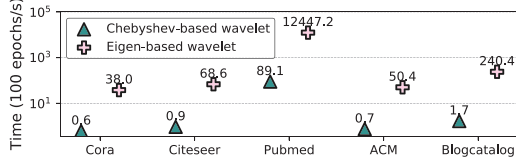
augmentation [21], [47]. However, the former often suffers from limited generalization capabilities, while the latter significantly increases training time and memory overhead. In contrast, FVD, as a plug-and-play method, strikes an effective balance between improving performance and maintaining training efficiency.

H. Discussion of Limitations

While FVD delivers substantial improvements across various VAE-based methods, several limitations need to be explained. Specifically, during the integration of FVD with CVGAE, we had to repeatedly adjust certain parameters within FVD. This issue arises because methods like CVGAE often require pre-training of specific modules, which can hinder the subsequent effective training of FVD. Similar issues arise in methods that



(a) Comparison of time costs between VGAE and VGAE+FVD on various datasets.



(b) Comparison of time costs between Chebyshev-approximated and eigenvalue-based wavelet transforms on various datasets.

Fig. 10. Runtime comparison.

TABLE VI
PERFORMANCE (%) COMPARISON OF VGAE+FVD USING
CHEBYSHEV-APPROXIMATED VS. EIGENVALUE-BASED WAVELET TRANSFORMS

Datasets	Metrics	VGAE+FVD	VGAE+FVD _{Eigen.}
Cora	ACC	67.21	68.57
	NMI	49.54	49.36
	F1	41.86	43.85
	ARI	66.55	67.21
Citeseer	ACC	51.85	52.72
	NMI	25.02	25.14
	F1	22.24	23.34
	ARI	51.02	51.39
Pubmed	ACC	64.97	66.03
	NMI	22.51	23.95
	F1	23.44	24.86
	ARI	64.68	65.93
ACM	ACC	84.99	85.85
	NMI	53.42	55.53
	F1	60.34	62.42
	ARI	85.04	85.84
Blogcatalog	ACC	45.28	48.52
	NMI	29.36	31.19
	F1	19.95	23.41
	ARI	43.89	47.03

rely on pre-training Gaussian mixture distributions, though these methods generally require only minor parameter adjustments. Notably, even vanilla DDPM encounters similar challenges in such scenarios. Additionally, we observe that FVD yields relatively modest performance improvements for certain methods and, in some instances, even minor performance declines. We attribute this to the differing impacts of different methods on signal amplitude, which may lead to blurred cluster boundary information. Specifically, some methods inherently suppress high-frequency signals or lack the capacity to accurately capture cluster boundaries. Consequently, the signal modulation introduced by the graph wavelet transform may inadvertently result in the loss of cluster boundary information.

VI. RELATED WORKS

A. Variational Autoencoder-Based Node Clustering

Graph machine learning has become a key interdisciplinary field for modeling complex relational data. For example, [8] represents a important effort at the intersection of drug discovery and graph learning, exploring molecular generation from

the perspective of graph motifs and significantly promoting the use of graph-based methods in real-world biomedical applications. [48] leverages large language models to efficiently and accurately construct cross-disciplinary knowledge graphs, effectively addressing the heavy reliance on manual intervention in existing methods. This significantly enhances the automation of knowledge extraction and integration, providing strong technical support for multidisciplinary knowledge fusion. [49] and [50] have also contributed to advancing research in graph machine learning. On this basis, node clustering plays a central role in understanding the structural organization of graphs. In recent years, VAEs have garnered widespread interest in node clustering tasks due to their strong ability to model complex data distributions. As mentioned before, existing VAE methods can be roughly divided into two categories: (1) extends the inference and/or generation processes by incorporating additional sample information, and (2) introduces constraints to modify the optimization function.

For the first category, VGAE as a representative method for unsupervised learning, has gradually become an important method for node clustering due to its competitive performance, interpretability and scalability. In recent years, numerous related variants have been developed to address specific challenges in graph representation and node clustering. For example, [38] observes that vanilla VGAE struggles to effectively encode isolated nodes. To address this, they propose incorporating L_2 normalization to generate more robust node embeddings. [40] introduces random walk-based regularization on the node representation, which effectively strengthens its capability to capture latent graph structures. [51] explore the use of Gaussian distributions in Wasserstein space as potential representations for nodes. Their method not only captures the graph structure but also incorporates node uncertainties, proving effective for handling real-world graphs with complex and irregular structures. Building on this foundation, [52] employs Wasserstein regularization directly on the latent space distribution to enhance the quality of node embeddings. Modularity is an effective metric for evaluating the quality of cluster partitions [53]. To leverage this, [7] introduce a modularity matrix during the inference process and reconstruct it during generation, aiming to enhance the accuracy of cluster partitions. [39] establishes a theoretical connection between variational Bayesian methods and graph partitioning by interpreting the reconstruction term in ELBO as a spectrally relaxed balanced graph cut. In addition to the aforementioned methods, related studies [9], [54], [55] summarize or provide new insights into node clustering tasks. Notably, these methods are not explicitly designed for node clustering, and thus often requiring downstream partitioning techniques, such as K-means, to identify cluster structures.

In contrast to the first category, methods in the second category are typically capable of performing end-to-end node clustering. Among them, one of the most representative approaches is proposed by [12], who first integrate Gaussian mixture models into the graph variational Bayesian framework. Each Gaussian distribution represents a cluster, enabling iterative optimization through a novel ELBO formulation to achieve end-to-end clustering. Expanding on this idea, [56] designs a simplified graph

convolutional encoder aimed at improving the convergence speed of VGAECD. Through the incorporation of dual variational objectives, their method achieves significant advancements in computational efficiency. Building on the contributions of these two works, a growing number of end-to-end node clustering methods have been developed, which incorporate Gaussian mixture models as a key component. For example, GM-VGAE [37] incorporates a Gaussian mixture model along with adversarial regularization to represent the prior distribution within the variational graph Bayesian framework. [11] observes that existing VAE methods fail to account for the differences between inference and generation models after incorporating clustering inductive bias. Moreover, they are poor at solving the problems of posterior collapse and attribute drift. To address these limitations, they integrate contrastive learning with the variational Bayesian framework and propose a more compact optimization objective to mitigate or resolve these challenges. Related studies [57], [58], [59], [60] have all contributed to advancing this research direction. For example, [61] proposes an attribute graph augmented clustering framework that explicitly models node relationships in both the topology and attribute spaces, thereby effectively improving clustering performance. Compared to methods that rely solely on explicit graph information, this approach further uncovers implicit correlations in the attribute space and provides new insights for model design. Notably, methods that integrate Gaussian mixture models typically require pre-training to initialize their parameters. Poorly chosen initial parameters can negatively impact the overall performance of these methods. Therefore, when integrating such methods with FVD, careful attention must be paid to tuning certain parameters in the FVD to ensure optimal results. Moreover, building on the concept of message invariance, [62] develops the topological compensation framework, which converts out-of-batch message passing into in-batch propagation. Demonstrated efficiency gains on million-node and billion-edge graphs mark a key step toward scalable graph machine learning, and this line of work has also inspired subsequent applications of variational autoencoder-based methods to large-scale graphs.

Beyond performance improvements, the structural consistency of representation spaces has also attracted increasing attention. Notably, DURA [63], as a representative work, introduces a duality-induced regularizer to address the limitations of inner-product-based attribute matching, ensuring that nodes with similar attributes are embedded closely. Inspired by this perspective, although FVD is not explicitly designed for attribute matching, its frequency-aware signal modulation may implicitly encourage more structured latent spaces, potentially enhancing the consistency of representations in downstream tasks such as node clustering.

B. Application of Diffusion Model on Graph

Denoising Diffusion Probabilistic Model [13] is a generative method that is widely applied in images, videos and texts, since it can learn the underlying data distribution and generate high-quality samples [64], [65]. In recent years, diffusion models have achieved remarkable success in the field of vision. Among

them, several studies combining diffusion models with VAEs have been proposed for image generation tasks [66]. Given the advantages of these methods over traditional generative models, research on diffusion models has gradually expanded and found applications in graph-based tasks. For example, [14] integrates diffusion models with the autoencoder framework to address the challenges of high resource costs in large-scale item prediction and temporal changes in user preferences within recommendation systems. [67] combines a graph diffusion model with a cross-modal contrastive learning paradigm to tackle the challenge of data sparsity in multi-modal recommendation tasks. [15] proposes integrating variational autoencoders with conditional diffusion models to improve the embeddings of historical neighbor nodes in continuous-time graphs. To better capture both local and global structures in topology and attribute spaces, [68] combines graph diffusion with contrastive learning and adaptive embedding fusion, yielding state-of-the-art performance on multiple benchmarks. Beyond the aforementioned methods, graph diffusion-based methods have also been applied to various graph tasks and applications, including anomaly detection [16], graph generation [69], [70], [71], protein structure and sequence generation [72], and link prediction [73]. It is worth noting that FVD fundamentally differs from these methods. Specifically, FVD is explicitly designed for node clustering and integrates principles from graph signal processing to align with the spectral characteristics of VAEs. To the best of our knowledge, no existing approach has attempted to apply conditional diffusion methods to VAEs for node clustering from the spectrum perspective.

VII. CONCLUSION

This paper explores why DDPM can enhance the expressiveness of the latent space in VAEs from the perspective of graph signal processing. Based on this insight, we conduct empirical studies and identify two key limitations of DDPM: inadequate high-frequency signal modulation and limited ability to effectively capture and reflect cluster characteristics in the latent space. To overcome these challenges, we propose FVD, a plug-and-play method that integrates a graph wavelet transform module and cluster-driven conditional diffusion modules. Comprehensive experiments demonstrate that baseline methods enhanced with FVD achieve competitive performance in node clustering tasks, while maintaining manageable computational overhead.

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